Peter Bhabra

Software Engineer | AI/ML & Distributed Systems

London, United Kingdom | LinkedIn

Software engineer with technical leadership experience in AI infrastructure and distributed systems. Currently leading inference platform development at Doubleword. Previously built high-availability data platforms processing 1M+ files daily at CMR Surgical.

Technologies: Rust, Python, SQL, TypeScript, React, PostgreSQL, Kubernetes, Docker, Helm, Terraform, AWS, GCP, Argo CD, GitHub Actions, vLLM, SGLang, Prometheus, Grafana, OpenTelemetry.

Specialisms: LLM inference, distributed systems, GPU scheduling, object storage, performance engineering, production operations.

EXPERIENCE

Doubleword | Member of Technical Staff - 2023 to Present

Built self-hosted inference products before moving to lead the core engineering team for Doubleword's hosted inference platform.

Hosted inference platform | Late 2025 - Present

- Led a five-person engineering team responsible for Doubleword's core hosted inference platform, and ran hiring processes to grow the team.
- Owned incident response and on-call coverage for a production inference service operating 24/7.
- Developed low-latency Rust backends for real-time and asynchronous inference, including an API gateway benchmarked at 450× faster than LiteLLM.
- Implemented durable request processing backed by PostgreSQL, covering dispatch, retries, completion, and failure handling.
- Extended NVIDIA Dynamo to serve LLMs across SGLang and vLLM backends, covering request routing, overflow to external providers, and inference metrics and tracing.

Self-hosted inference | 2023 - Late 2025

- Built custom inference engines with scheduling and resource management, and implemented inference research including speculative decoding.
- Delivered full-stack, on-premises deployments of open-source LLMs, production APIs, and RAG applications.
- Led technical client engagements throughout B2B sales cycles and deployment.
- Built a Rust-based inference engine using Candle for Stable Diffusion models.
- Built release and deployment automation, Helm charts, and custom Kubernetes operators, and managed self-hosted CI/CD clusters.

CMR Surgical | Graduate Software Engineer - 2021 to 2023

Built and maintained AWS data platforms processing over one million files daily. Reduced monthly AWS costs from $60,000 to $35,000 (42%) within one month.

EDUCATION

Durham University | Master of Science · Scientific Computing and Data Analysis - First Class Honours

Durham University | Bachelor of Science · Physics - First Class Honours

SELECTED TALKS

The Massively Parallel Agent Stack - Cloud Native & Open Source AI Conference · 2026

Gateways Unlock AI Builders - Cloud Native AI UG · 2025

TECHNICAL WRITING

Writing on AI infrastructure, systems engineering, and performance. Read the blog

Gigatoken: making tokenisation care about the hardware